Fiscal Year:	FY 2013	**Task Last Updated:**	FY 05/14/2013
PI Name:	Trivedi, Rohit Ph.D.		
Project Title:	Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	06/12/2012	**End Date:**	06/11/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:	May 2013 report: Dr. Jiho Gu, a post-doc scientist completed his term and returned to S. Korea.		
COI Name (Institution):	Karma, Alain (Northeastern University)		
Grant/Contract No.:	NNX12AK54G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The proposed research builds on insights gained through analysis of flight experimental data from the previous DSI investigation as well as associated computational simulation results, while identifying distinct research objectives for the new investigation. The proposed research involves solidification of organic alloys used as a model of metallic alloy systems. It focuses on the origin and development of the dendritic sidebranch instability that is crucial for dendrite growth and for determining the solute segregation pattern in the mushy zone that largely governs the properties of cast alloys. The primary goal of the proposed research is to quantitatively establish the fundamental physics controlling the spatiotemporal organization of the secondary sidebranch structure and its interaction with the primary array structure. Ground based experiments will be conducted to determine the appropriate sample composition and the experimental matrix of growth conditions (thermal gradient and solidification velocity) for the flight experiments. The proposed research will also involve phase field simulations of the proposed growth conditions to aid in determination of the		

	experimental matrix. These simulation results will also be used to validate the model using the flight experimental data.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. By bringing together well-controlled and accurately measured three-dimensional solidification experiments with quantitative phase-field simulations, this research should bring significant progress in understanding the origin of side-branching and the nature of the cell to dendrite transition, which have been a long standing issues in solidification for several decades. The deeper understanding of microstructure formation dynamics shall thus accelerate and optimize the design of new technological alloys that can directly benefit U.S. industries.
Task Progress:	Since the DECLIC hardware was brought to ground, all the work during this period was ground-based. These ground-based studies were carried out for three different aspects of the project. (1) Analysis of the experimental data obtained from the International Space Station. (2) Development of the rigorous theoretical model and application of this model to the steady-state and oscillatory growth of cells and dendrites. (3) Ground-based experiments under diffusive growth conditions in thin samples to determine the composition and the experimental matrix for the next set of experiments to be carried out on the International Space Station. In the cellular regime, microgravity experiments have revealed a rich variety of 3D breathing modes with limited spatiotemporal coherence. At low growth velocities and high enough temperature gradient, these breathing modes exhibit a global oscillatory behavior with an oscillation period homogeneous over the spatially extended array. However, phase locking was not found to occur throughout the whole system, but only very locally among neighboring cells. These experimental observations were reproduced with phase-field simulations with a remarkable accuracy on the predicted oscillation period (e.g. experiments: t ˜ 45.6 min, phase-field: t ˜ 44.6 min, at G=2.8K/mm and V=1µm/s). Moreover, simulations explained the underlying physics of this oscillatory behavior: (i) phase-field modeling has shown that under some velocity and thermal gradient conditions no steady-state solutions occur over a narrow range of spacing, and the oscillatory behavior is found to occur at the end of a stable spacing branch before a gap in the stable array spacing is predicted; (ii) the lack of global coherence in experiments is due to the spatial reorganization of cells, maintained notably by tip splitting events. The identification of this subtle oscillatory mechanism constitutes the first quantitative agreement between observed and calculated dynamics in a three-dimensional configuration. The oscillatory behavior was observed over a range of velocity and the period of oscillation was determined as a function of velocity, and a scaling law was found as V-3/2. In the dendritic regime, experiments on a SCN - 0.24 wt% camphor alloy suggested the existence of possible coherent modes of dendritic sidebranching for high growth velocities. This mechanism of coherent side branching was earlier reported in thin sample experiments on a dilute SCN alloy as well as in 2D and thin-sample 3D phase-field simulations, showing that this mechanism of coherent branching events is related to a limit cycle. We confirmed the existence of this coherent branching mechanism with full 3D phase-field simulations. Simulations are now being performed on the SCN - camphor alloys under the microgravity experiment conditions leading to coherent side branching events. However, dendrites were only observed in a high velocity regime where the solidification front exhibited significant curvature due to the combination of the rejection of latent heat and the much lower thermal conductivity of the alloy than its quartz container wall. In addition, dendrite tip radii at these high velocities were found to be too small to be accurately resolved. We expect the next set of experiments to address these limitations by selecting appropriate composition and growth conditions. One of the key issues for the next set of experiments on the ISS is to determine the composition of the alloy to be used that will provide dendritic structures that could be analyzed accurately. Directional solidification runs in thin samples were carried out for different compositions ranging from 0.1 to 0.9 wt% camphor. Since earlier studies on ISS were carried out in succinonitrile-0.24 wt% camphor, we concentrated these studies on a lower composition of 0.1 wt% camphor and a higher composition of 0.5 wt% camphor. Detailed studies on the morphology, primary spacing, and tip radius were carried out for each of these compositions over a velocity range of 1 to 20 µm/s, and a thermal gradient range of 1.4 to 4.5 K/mm. The lower composition was examined since the critical velocity will be larger so that cellular growth can be examined just above the critical velocity where the tip radius will be larger and can be measured accurately by the laser interference in the DECLIC facility. The higher composition was examined since the critical velocity will be lower and dendrites will form at lower velocity. In a 0.1 wt % camphor alloy the microstructure was observed to be cellular for all thermal gradient examined for velocities up to 20 µm/s, so that a much higher velocities would be required to examine dendritic structures. As we observed in the earlier set of experiments on the International Space Station, the macroscopic interface at higher velocities exhibits significant curvature. In addition, dendrite tip radii at these higher velocities are too small to be accurately resolved. Thus, 0.1 wt% camphor alloy is found to be unsuitable for studying the dendrite side-branch formation. Experimental studies in succinonitrile-0.5 wt% camphor showed dendrite formation between velocities of 5.0 and 20.0 µm/s at thermal gradients of 1.5 to 3.0 that were predicted to be present in the earlier set of experiments on ISS. The formation of dendrites at a lower velocity also gives a higher tip radius that could be directly measure by ana;lyziing interferometry results. As a check on this alloy composition, detailed phase-field simulations were carried out for different G and V, and excellent agreement was observed on the microstructure. Also, measured and simulated dendrite tip radius showed very good quantitative agreement and tip radii were sufficiently large and they could be directly and accurately measured by interferometry. In addition, we also observed the formation of doublets, which would be multiplets in 3D, over a velocity range of 4.23 to 6.35 µm/s at G =3.15 K/mm, so that the formation of multiplets can also be examined in this alloy composition. Consequently, the alloy of succononitrile-0.5 wt% camphor was selected for the next set of experiments on the International Space Station. A sample of high purity succinonitrile, mixed with 0.5 wt% camphor, was prepared and inserted into the Directional Solidification Insert (DSI).
Bibliography Type:	Description: (Last Updated: 04/03/2024)

Abstracts for Journals and Proceedings	Billia B, Bergeon N, Chen L, Ramirez A, Trivedi R, Karma A, Tourret D, Debierre JM, Guerin R. "Dynamical formation of three-dimensional cellular and dendritic microstructures in alloy directional solidification under diffusion transport and under gravity-driven fluid flow." MS&T '12: Materials Science & Technology 2012 Conference & Exhibition, Pittsburgh, PA, October 7-11, 2012. Symposium Crystal Growth and Microstructural Correlation with Properties of Materials. MS&T '12: Materials Science & Technology 2012 Conference & Exhibition, Pittsburgh, PA, October 7-11, 2012. , Oct-2012
Abstracts for Journals and Proceedings	Bergeon N, Billia B, Chen L, Ramirez A, Trivedi R, Gu J, Karma A, Tourret D, Debierre JM, Guerin R. "Three-dimensional cellular and dendritic patterns under diffusion transport: In situ characterization of growth dynamics in DECLIC-Directional Solidification Insert onboard ISS." IAC 63: 63rd International Astronautical Congress, Naples, Italy, October 1-5, 2012. IAC 63: 63rd International Astronautical Congress, Naples, Italy, October 1-5, 2012. , Oct-2012
Abstracts for Journals and Proceedings	Bergeon N, Trivedi R, Chen L, Ramirez A, Billia B, Debierre JM, Gu J, Tourret D, Karma A. "Dynamical evolution and stability of spatially extended three dimensional interface patterns in directional solidification of alloys under diffusive growth conditions in microgravity." ICSPS-5: Fifth International Conference on Solidification Science and Processing, Bhubaneswar, India, November 19-22, 2012. ICSPS-5: Fifth International Conference on Solidification Science and Processing, Bhubaneswar, India, November 19-22, 2012. , Nov-2012
Abstracts for Journals and Proceedings	Bergeon N, Trivedi R, Billia B, Chen L, Gu J, Karma A, Tourret D, Debierre JM, Guerin R. "Dynamical evolution of three-dimensional interface patterns in directional solidification of alloys: Results from International Space Station Experiments." 1st Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 26-28, 2012. 1st Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 26-28, 2012. , Jun-2012
Articles in Peer-reviewed Journals	Bergeon N, Tourret D, Chen L,Debierre JM, Guerin R, Ramirez A, Billia B, Karma A, Trivedi R. "Spatiotemporal dynamics of oscillatory cellular patterns in three-dimensional directional solidification" Physical Review Letters, Accepted for publication, as of May 2013. http://prl.aps.org/accepted/9107bYc9O2112642f11864e52aed62b488aec7598 , May-2013